SOC Estimation: Coulomb Counting, KF, and EKF

This document explains the three estimators in this project, derives their equations, and works through numerical examples. It describes the current Python implementations rather than a general-purpose BMS with every possible feature.

The examples use illustrative values to explain the arithmetic. They are not measured cell parameters, benchmark results, or evidence that one estimator is more accurate than another.

1 Definitions and Conventions

State of charge (SOC) is the fraction of the chosen battery capacity remaining:

$$SOC = \frac{\text{remaining charge}}{\text{reference capacity}}$$

In these implementations, SOC is a fraction between 0 and 1. For example, 0.80 means 80% SOC. Multiply by 100 only when displaying a percentage.

Symbol	Meaning	Units
z	SOC; a shorter name used in the filter equations	Fraction
$\hat{z}$	Estimated SOC	Fraction
Q_{Ah}	Capacity used by the estimator	Amp-hours (Ah)
I_k	Measured current at sample k	Amperes (A)
Δt_k	Time since the previous sample	Seconds (s)
$\bar{I}_k$	Average of the two neighboring current samples	Amperes (A)
V_k^{meas}	Measured terminal voltage	Volts (V)
$OCV(z)$	Open-circuit voltage as a function of SOC	Volts (V)
R_0, R_1	Battery-model resistances	Ohms (Ω)
C_1	Battery-model polarization capacitance	Farads (F)
v_p	Polarization voltage	Volts (V)
P	State-error variance or covariance	Depends on the state
$\mathbf{Q}_k$	Process-noise covariance for one interval	Depends on the state
R_V	Voltage-measurement noise variance	V^2

Capacity Q_{Ah} and process-noise covariance $\mathbf{Q}_k$ are different quantities. The notation distinguishes them deliberately.

The current convention is:

- Positive current means discharge and decreases SOC.
- Negative current means charging and increases SOC.
- Current equal to zero removes no charge in these models.

Each estimator uses the actual timestamps:

$$\Delta t_k = t_k - t_{k-1}$$

Each also uses the average of adjacent current measurements:

$$\bar{I}_k = \frac{I_{k-1} + I_k}{2}$$

This is trapezoidal integration. It is exact for a linearly changing current within an interval, but a switched current step needs its timing represented correctly. Two endpoint readings alone do not tell us when the switch occurred.

1.1 Why OCV Appears in the Filters

Open-circuit voltage is the voltage of a cell with no current flowing after sufficient relaxation. Terminal voltage under load also includes resistance and polarization effects. The one-RC model expresses their relationship as:

$$V_{\text{terminal}} = OCV(z) - IR_0 - v_p$$

An OCV–SOC relationship lets a filter turn its SOC estimate into a voltage prediction. Comparing that prediction with the measured voltage provides information for correcting SOC. The project KF uses a straight-line approximation; the EKF accepts a nonlinear OCV function. These relationships follow the standard equivalent-circuit formulation described in the MathWorks SOC estimator equations.

2 Coulomb Counting

Implementation: `python/estimation/coulomb_counter.py`.

2.1 What It Does

Coulomb counting tracks charge entering or leaving the battery by integrating current. It needs a starting SOC and a capacity. It does not use voltage to correct the estimate or maintain a probabilistic uncertainty estimate.

Its basic update for a constant current over one interval is:

$$SOC_{\text{new}} = SOC_{\text{old}} - \frac{I\Delta t}{3600 Q_{\text{Ah}}}$$

2.2 Derivation: Where the Equation Comes From

Current is charge flow per second. For constant current, the removed charge is:

$$\Delta q_{\mathrm{C}} = I\Delta t$$

One amp-hour is 3600 coulombs, so the same removed charge in amp-hours is:

$$\Delta q_{\mathrm{Ah}} = \frac{I\Delta t}{3600}$$

Divide removed charge by battery capacity to obtain the removed SOC fraction:

$$\Delta SOC = \frac{\Delta q_{\mathrm{Ah}}}{Q_{\mathrm{Ah}}} = \frac{I\Delta t}{3600 Q_{\mathrm{Ah}}}$$

Subtract that fraction because positive current means discharge. For a varying current, the continuous equation is:

$$z(t) = z(t_0) - \frac{1}{3600 Q_{\mathrm{Ah}}} \int_{t_0}^{t} I(\tau)\, d\tau$$

Approximating the integral over each interval with a trapezoid gives the equation in the current code:

$$\hat{z}_k = \hat{z}_{k-1} - \frac{\bar{I}_k \Delta t_k}{3600 Q_{\mathrm{Ah}}}$$

2.3 Worked Example

Assume SOC starts at 80%, capacity is 2 Ah, and discharge current stays at 1 A for 60 seconds.

$$\Delta q_{\mathrm{Ah}} = \frac{1 \times 60}{3600} = 0.0166667 \text{ Ah}$$

$$\Delta SOC = \frac{0.0166667}{2} \approx 0.0083333$$

$$SOC_{\mathrm{new}} = 0.80 - 0.0083333 \approx 0.7916667 = 79.16667\%$$

The decrease is about 0.83333 percentage points. A current of -1 A over the same interval would instead increase SOC to about 80.83333%, assuming the same charge efficiency.

2.4 Why Current Offset Causes Drift

Suppose measured current has a constant offset b_I. With the same capacity and away from clipping, SOC error follows:

$$e_z(t) = e_z(t_0) - \frac{b_I(t - t_0)}{3600 Q_{\mathrm{Ah}}}$$

For a $+0.010$ A offset over one hour on a 2 Ah battery:

$$\Delta e_z = -\frac{0.010 \times 3600}{3600 \times 2} = -0.005$$

That is an additional -0.5 percentage-point SOC error. An incorrect starting SOC also persists until clipping affects the result.

2.5 Classification and Limitations

With fixed capacity and timestamps, the underlying charge integration is linear in current and initial SOC. It is a deterministic integration method, not a Kalman filter.

The implementation assumes unit charge efficiency and does not explicitly model self-discharge or capacity changes. It clips SOC to 0–1 after each interval. Clipping is nonlinear and discards accumulated error outside that range, so it can obscure drift near empty or full SOC.

3 Linear Kalman Filter (KF)

Implementation: `python/estimation/kalman_filter.py`.

3.1 What It Does

The KF combines a current-based SOC prediction with a voltage-based correction. It maintains uncertainty about SOC and uses that uncertainty to decide how strongly the voltage measurement should influence the result.

This project's KF has one state: SOC. Its voltage model is:

$$OCV(z) \approx b + mz$$

$$b = OCV(0), \qquad m = OCV(1) - OCV(0)$$

$$V = b + mz - IR_0$$

The measurement relationship is affine: a straight line plus known offsets. Subtracting the known offsets gives the standard linear measurement form. There is no separate polarization state in this KF.

3.2 Step 1: Predict SOC

The prediction is Coulomb counting:

$$\hat{z}_k^- = \hat{z}_{k-1}^+ - \frac{\bar{I}_k \Delta t_k}{3600 Q_{\mathrm{Ah}}}$$

The superscript minus means before using the current voltage measurement; plus means after correction. They do not indicate negative or positive SOC.

Because the sensitivity to the previous SOC is 1, the scalar state-transition factor is $F = 1$. The current implementation adds a configured process-variance rate q_z over the elapsed time:

$$P_k^- = F P_{k-1}^+ F^T + q_z \Delta t_k = P_{k-1}^+ + q_z \Delta t_k$$

3.3 Step 2: Predict Voltage and Calculate the Residual

$$\hat{V}_k = b + m\hat{z}_k^- - I_k R_0$$

$$r_k = V_k^{\mathrm{meas}} - \hat{V}_k$$

The voltage sensitivity to SOC is $H = m$. Therefore, the predicted variance of the voltage residual is:

$$S_k = HP_k^- H^T + R_V = m^2 P_k^- + R_V$$

Variance is the square of standard deviation. For example, a voltage standard deviation of 0.020 V corresponds to $R_V = 0.0004\ \text{V}^2$, not 0.020.

3.4 Step 3: Calculate the Gain and Correct SOC

$$K_k = \frac{P_k^- m}{m^2 P_k^- + R_V}$$

$$\hat{z}_k^+ = \hat{z}_k^- + K_k r_k$$

The gain converts a voltage residual into an SOC correction. It is not a dimensionless percentage weight: here its units are SOC fraction per volt.

For the same model and prior, a larger measurement variance reduces reliance on voltage. A larger predicted SOC variance increases its influence. The code updates uncertainty with scalar Joseph form:

$$P_k^+ = (1 - K_k m)^2 P_k^- + K_k^2 R_V$$

3.5 Worked Example

Consider an update beginning with SOC 0.80 and posterior variance 0.0025. These values describe the start of this illustrative interval, not necessarily the first iteration with the code's defaults.

Use the following values:

- Capacity: 2 Ah; current: constant 1 A; elapsed time: 60 s.
- $b = 3.0$ V, $m = 1.2$ V per SOC fraction, and $R_0 = 0.05\ \Omega$.
- $q_z = 10^{-8}$ per second and $R_V = 0.020^2 = 0.0004\ \text{V}^2$.
- Measured terminal voltage: 3.920 V.

First, integrate current and propagate uncertainty:

$$\hat{z}_k^- = 0.80 - \frac{1 \times 60}{3600 \times 2} \approx 0.7916667$$

$$P_k^- = 0.0025 + 10^{-8} \times 60 = 0.0025006$$

Then predict voltage and compare it with the measurement:

$$\hat{V}_k = 3.0 + 1.2(0.7916667) - 1(0.05) \approx 3.900\ \text{V}$$

$$r_k = 3.920 - 3.900 = 0.020\ \text{V}$$

Calculate residual variance and gain:

$$S_k = 1.2^2(0.0025006) + 0.0004 = 0.004000864$$

$$K_k = \frac{0.0025006(1.2)}{0.004000864} \approx 0.750018$$

Finally, correct SOC and its variance:

$$\hat{z}_k^+ \approx 0.7916667 + 0.750018(0.020) \approx 0.8066670 = 80.66670\%$$

$$P_k^+ = (1 - 0.750018 \times 1.2)^2(0.0025006) + (0.750018)^2(0.0004) \approx 0.000250006$$

The measured voltage was higher than predicted, so this model increased SOC. This is a correction to the estimate, not evidence that charge entered the battery.

3.6 Classification and Limitations

This is a linear-model Kalman filter before the code's SOC clipping. Its main simplifications are the straight-line OCV curve, fixed capacity, and single resistance. It cannot separately attribute voltage changes to polarization, so model error may be interpreted as SOC error.

The Kalman equations can be implemented correctly while the SOC result is inaccurate because the voltage model or uncertainty settings are inappropriate.

4 Extended Kalman Filter (EKF)

Implementation: `python/estimation/ekf.py`.

4.1 What It Does

The EKF retains current integration but adds a nonlinear OCV function and a polarization-voltage state:

$$\mathbf{x}_k = \begin{bmatrix} z_k \\ v_{p,k} \end{bmatrix}$$

The nonlinear measurement model is:

$$h(\mathbf{x}_k, I_k) = OCV(z_k) - I_k R_0 - v_{p,k}$$

The OCV function is nonlinear when its slope changes with SOC. The EKF evaluates the full function for voltage prediction and uses its local slope to calculate the correction.

4.2 Derive the Polarization Dynamics

In the parallel R_1–C_1 branch, current divides between the resistor and capacitor:

$$I = \frac{v_p}{R_1} + C_1 \frac{dv_p}{dt}$$

Rearrange:

$$\frac{dv_p}{dt} = \frac{I}{C_1} - \frac{v_p}{R_1 C_1}$$

Define the time constant $\tau = R_1 C_1$. With constant current over an interval, solving this differential equation gives:

$$v_p(t + \Delta t) = IR_1 + [v_p(t) - IR_1]\, e^{-\Delta t/\tau}$$

Define $a_k = e^{-\Delta t_k/(R_1 C_1)}$ and rearrange:

$$\hat{v}_{p,k}^{-} = a_k \hat{v}_{p,k-1}^{+} + R_1(1 - a_k)\bar{I}_k$$

This is the project's update: it treats the interval-average current as constant. The solution is exact for constant current, but approximate when current varies within the interval. During rest, the forcing term is zero and polarization decays exponentially.

4.3 Step 1: Predict Both States

$$\hat{\mathbf{x}}_k^{-} = \begin{bmatrix} \hat{z}_{k-1}^{+} - \dfrac{\bar{I}_k \Delta t_k}{3600 Q_{\text{Ah}}} \\ a_k \hat{v}_{p,k-1}^{+} + R_1(1 - a_k)\bar{I}_k \end{bmatrix}$$

With fixed circuit parameters, differentiation with respect to the previous states gives the transition Jacobian:

$$\mathbf{F}_k = \begin{bmatrix} 1 & 0 \\ 0 & a_k \end{bmatrix}$$

These entries describe sensitivity: a change in previous SOC passes through unchanged, while a change in previous polarization is multiplied by a_k.

The code uses a diagonal process covariance built from configured variance rates:

$$\mathbf{Q}_k = \begin{bmatrix} q_z \Delta t_k & 0 \\ 0 & q_p \Delta t_k \end{bmatrix}$$

$$\mathbf{P}_k^{-} = \mathbf{F}_k \mathbf{P}_{k-1}^{+} \mathbf{F}_k^{T} + \mathbf{Q}_k$$

The diagonal entries of $\mathbf{P}$ are SOC variance and polarization-voltage variance. Its off-diagonal entries describe their error correlation. The linear-in-time process covariance is a modeling choice; it is not a full derivation from every physical noise source.

4.4 Step 2: Predict Voltage and Linearize the Observation

$$\hat{V}_k = OCV(\hat{z}_k^{-}) - I_k R_0 - \hat{v}_{p,k}^{-}$$

$$r_k = V_k^{\text{meas}} - \hat{V}_k$$

A first-order expansion around the predicted state is:

$$h(\hat{\mathbf{x}}_k^{-} + \delta\mathbf{x}, I_k) \approx h(\hat{\mathbf{x}}_k^{-}, I_k) + \mathbf{H}_k \delta\mathbf{x}$$

The measurement Jacobian follows by differentiating the voltage model:

$$\mathbf{H}_k = \begin{bmatrix} \left.\dfrac{dOCV}{dz}\right|_{\hat{z}_k^{-}} & -1 \end{bmatrix}$$

The second entry is -1 because increasing polarization decreases terminal voltage. The code accepts an OCV derivative function or approximates it numerically, using one-sided differences near the SOC boundaries.

4.5 Step 3: Calculate the Gain and Correct Both States

There is one voltage measurement, so residual variance S_k is a scalar even though the state has two entries:

$$S_k = \mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + R_V$$

$$\mathbf{K}_k = \frac{\mathbf{P}_k^- \mathbf{H}_k^T}{S_k}$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k r_k$$

The gain has two entries. One changes SOC; the other changes polarization voltage. The code uses the matrix Joseph covariance update, also used in the FilterPy EKF implementation:

$$\mathbf{P}_k^+ = (\mathbf{I}_2 - \mathbf{K}_k \mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I}_2 - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k R_V \mathbf{K}_k^T$$

Here $\mathbf{I}_2$ is the two-by-two identity matrix, not battery current. Joseph form helps preserve a valid covariance under floating-point rounding.

4.6 Worked Example

This is a separate illustrative model from the KF example. It demonstrates one update, not a comparison of the two filters' performance.

Suppose an interval starts with:

$$\hat{\mathbf{x}}_{k-1}^+ = \begin{bmatrix} 0.80 \\ 0 \end{bmatrix}, \qquad \mathbf{P}_{k-1}^+ = \begin{bmatrix} 0.0025 & 0 \\ 0 & 0.0001 \end{bmatrix}$$

Use capacity 2 Ah, constant current 1 A, elapsed time 10 s, $R_0 = 0.05\ \Omega$, $R_1 = 0.02\ \Omega$, and $C_1 = 500$ F. Let $q_z = 10^{-8}$, $q_p = 10^{-6}$, and $R_V = 0.0004\ \text{V}^2$, with variance rates expressed per second in their respective units.

For this example only, define:

$$OCV(z) = 3.0 + 0.8z + 0.4z^2, \qquad \frac{dOCV}{dz} = 0.8 + 0.8z$$

First calculate the time constant and decay:

$$\tau = 0.02 \times 500 = 10\ \text{s}, \qquad a_k = e^{-10/10} \approx 0.3678794$$

Predict SOC and polarization:

$$\hat{z}_k^- = 0.80 - \frac{1 \times 10}{3600 \times 2} \approx 0.7986111$$

$$\hat{v}_{p,k}^- = 0.3678794(0) + 0.02(1 - 0.3678794)(1) \approx 0.01264241\ \text{V}$$

Propagate covariance. Since the starting matrix is diagonal, this multiplication is especially simple:

$$\mathbf{P}_k^- = \begin{bmatrix} 0.0025 + 10^{-8}(10) & 0 \\ 0 & (0.3678794)^2(0.0001) + 10^{-6}(10) \end{bmatrix} \approx \begin{bmatrix} 0.0025001 & 0 \\ 0 & 0.00002353353 \end{bmatrix}$$

Predict terminal voltage:

$$OCV(0.7986111) \approx 3.89400077 \text{ V}$$

$$\hat{V}_k \approx 3.89400077 - 1(0.05) - 0.01264241 \approx 3.83135836 \text{ V}$$

Suppose measured voltage is 3.85135836 V. The residual is approximately +0.020 V. Evaluate the local OCV slope:

$$\mathbf{H}_k = \begin{bmatrix} 0.8 + 0.8(0.7986111) & -1 \end{bmatrix} \approx \begin{bmatrix} 1.4388889 & -1 \end{bmatrix}$$

Calculate residual variance and the two gains:

$$S_k \approx (1.4388889)^2(0.0025001) + (-1)^2(0.00002353353) + 0.0004 \approx 0.00559974365$$

$$\mathbf{K}_k \approx \frac{1}{0.00559974365} \begin{bmatrix} 0.0025001(1.4388889) \\ 0.00002353353(-1) \end{bmatrix} \approx \begin{bmatrix} 0.64241621 \\ -0.004202608 \end{bmatrix}$$

Apply the correction:

$$\hat{\mathbf{x}}_k^+ \approx \begin{bmatrix} 0.7986111 \\ 0.01264241 \end{bmatrix} + \begin{bmatrix} 0.64241621 \\ -0.004202608 \end{bmatrix} (0.020) \approx \begin{bmatrix} 0.81145944 \\ 0.01255836 \end{bmatrix}$$

The corrected SOC is about 81.14594%. The filter also decreases polarization slightly. Both changes raise predicted voltage, consistent with the positive residual.

Using the Joseph update gives:

$$\mathbf{P}_k^+ \approx \begin{bmatrix} 0.00018909369 & 0.00001511832 \\ 0.00001511832 & 0.00002343463 \end{bmatrix}$$

The new off-diagonal entries show that the two state errors are now correlated because the same voltage observation corrected both states. Intermediate calculations above are rounded; use full precision when implementing the equations.

4.7 Classification and Limitations

The EKF is used because the observation model is nonlinear. With fixed capacity and RC parameters, the state prediction itself is affine before clipping. Each correction uses a local linear approximation to the nonlinear voltage model. The standard one-RC model and Jacobians are documented in the MathWorks SOC estimator equations.

A small OCV slope gives voltage less information about SOC. Circuit parameters, OCV calibration, initial uncertainty, and noise settings all affect performance. The EKF does not guarantee better accuracy just because its model is more detailed.

5 How the Methods Compare in This Project

Property	Coulomb counting	Linear KF	EKF
Uses current integration	Yes	Yes	Yes
Uses measured voltage for correction	No	Yes	Yes
Estimated states	SOC	SOC	SOC and polarization voltage
OCV model	None	Straight line	Supplied nonlinear function or lookup
Tracks uncertainty	No	Scalar variance	Two-by-two covariance
Can recover from wrong initial SOC	No inherent correction	If voltage model and tuning support it	If model, observability, and tuning support it
Main role	Simple baseline	Simplified model-based baseline	More detailed model-based estimator

The distinction is not simply three unrelated algorithms: the filters both use Coulomb counting for SOC prediction, then add voltage-based correction.

6 Details Specific to the Current Code

- At the first sample, Coulomb counting returns the supplied initial SOC. KF and EKF immediately use the first voltage observation to correct their initial states; they perform no time integration at that first sample.
- All three accept unequal time intervals and reject backward timestamps. They allow equal timestamps. Independent measurements at the same time can be valid, but accidentally replaying the same voltage record causes an extra filter correction.
- KF and EKF treat measured current as a known input and use configured process variance to represent prediction uncertainty. They do not explicitly estimate current bias or model all correlations caused by current-measurement noise.
- Current integration assumes unit charge efficiency and fixed capacity. Temperature is not an estimator input in these three functions.
- All three clip SOC to 0–1. EKF clips both its SOC prediction and corrected SOC; KF clips corrected SOC. The covariance updates do not account for those constraints, and the displayed Jacobians describe the unconstrained interior model.
- Reference SOC belongs in evaluation, not in the estimator correction. In a hardware experiment, a Coulomb-count reference also has uncertainty.
- The EKF requires an OCV function and R_0, R_1, C_1 from its caller. The current estimator files do not calibrate those values themselves.

The first calculation to reproduce by hand is the Coulomb-counting example. Next, reproduce the scalar KF's prediction and voltage correction. The EKF adds a second state, a nonlinear OCV function, and matrix uncertainty propagation to that same prediction-and-correction idea.